



Volume and Dimensional Scaling

Grades 9-12 • SAT / ACT mathematics foundations

Geometry and Trigonometry • Use volume formulas and cubic scale factors.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Use volume formulas and cubic scale factors.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. A rectangular prism measures 4 cm by 5 cm by 8 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

Original volume = $4 \times 5 \times 8 = 160$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 4320 cm^3 .

Model 2. A rectangular prism measures 5 cm by 9 cm by 6 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

Original volume = $5 \times 9 \times 6 = 270$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 7290 cm^3 .

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A rectangular prism measures 3 cm by 2 cm by 3 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

2. A rectangular prism measures 8 cm by 8 cm by 4 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

3. A rectangular prism measures 4 cm by 3 cm by 5 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A rectangular prism measures 6 cm by 4 cm by 6 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

5. A rectangular prism measures 8 cm by 6 cm by 2 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

6. A rectangular prism measures 8 cm by 5 cm by 5 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A rectangular prism measures 4 cm by 3 cm by 5 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

8. A rectangular prism measures 5 cm by 6 cm by 6 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

9. A rectangular prism measures 9 cm by 3 cm by 8 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A rectangular prism measures 8 cm by 8 cm by 2 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

11. A rectangular prism measures 7 cm by 8 cm by 2 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

12. A rectangular prism measures 8 cm by 3 cm by 6 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A rectangular prism measures 4 cm by 6 cm by 3 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

14. A rectangular prism measures 9 cm by 6 cm by 7 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

15. A rectangular prism measures 8 cm by 7 cm by 7 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A rectangular prism measures 7 cm by 8 cm by 6 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.
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Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A rectangular prism measures 2 cm by 7 cm by 7 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

2. A rectangular prism measures 5 cm by 3 cm by 5 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A rectangular prism measures 9 cm by 9 cm by 7 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

4. A rectangular prism measures 6 cm by 5 cm by 8 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A rectangular prism measures 5 cm by 8 cm by 4 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

6. A rectangular prism measures 7 cm by 8 cm by 5 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A rectangular prism measures 8 cm by 8 cm by 5 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

2. A rectangular prism measures 2 cm by 6 cm by 7 cm. Every dimension is multiplied by 3. Find the new volume and compare it with the original.



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A rectangular prism measures 7 cm by 2 cm by 8 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.

4. A rectangular prism measures 3 cm by 3 cm by 4 cm. Every dimension is multiplied by 2. Find the new volume and compare it with the original.



Practice worked solutions

1. 486 cm³; 27 times as large

Original volume = $3 \times 2 \times 3 = 18$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$;
new volume = 486 cm³.

2. 6912 cm³; 27 times as large

Original volume = $8 \times 8 \times 4 = 256$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$;
new volume = 6912 cm³.

3. 1620 cm³; 27 times as large

Original volume = $4 \times 3 \times 5 = 60$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$;
new volume = 1620 cm³.

4. 1152 cm³; 8 times as large

Original volume = $6 \times 4 \times 6 = 144$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$;
new volume = 1152 cm³.

Practice worked solutions

5. 768 cm³; 8 times as large

Original volume = $8 \times 6 \times 2 = 96$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 768 cm³.

6. 1600 cm³; 8 times as large

Original volume = $8 \times 5 \times 5 = 200$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 1600 cm³.

7. 1620 cm³; 27 times as large

Original volume = $4 \times 3 \times 5 = 60$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 1620 cm³.

8. 4860 cm³; 27 times as large

Original volume = $5 \times 6 \times 6 = 180$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 4860 cm³.

Practice worked solutions

9. 5832 cm³; 27 times as large

Original volume = $9 \times 3 \times 8 = 216$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$;
new volume = 5832 cm³.

10. 1024 cm³; 8 times as large

Original volume = $8 \times 8 \times 2 = 128$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$;
new volume = 1024 cm³.

11. 3024 cm³; 27 times as large

Original volume = $7 \times 8 \times 2 = 112$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$;
new volume = 3024 cm³.

12. 3888 cm³; 27 times as large

Original volume = $8 \times 3 \times 6 = 144$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$;
new volume = 3888 cm³.

Practice worked solutions

13. 576 cm³; 8 times as large

Original volume = $4 \times 6 \times 3 = 72$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 576 cm³.

14. 10206 cm³; 27 times as large

Original volume = $9 \times 6 \times 7 = 378$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 10206 cm³.

15. 3136 cm³; 8 times as large

Original volume = $8 \times 7 \times 7 = 392$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 3136 cm³.

16. 2688 cm³; 8 times as large

Original volume = $7 \times 8 \times 6 = 336$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 2688 cm³.



Transfer worked solutions

1. 2646 cm³; 27 times as large

Original volume = $2 \times 7 \times 7 = 98$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 2646 cm³.

2. 600 cm³; 8 times as large

Original volume = $5 \times 3 \times 5 = 75$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 600 cm³.

3. 4536 cm³; 8 times as large

Original volume = $9 \times 9 \times 7 = 567$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 4536 cm³.

4. 6480 cm³; 27 times as large

Original volume = $6 \times 5 \times 8 = 240$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 6480 cm³.

Transfer worked solutions

5. 1280 cm³; 8 times as large

Original volume = $5 \times 8 \times 4 = 160$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$;
new volume = 1280 cm³.

6. 2240 cm³; 8 times as large

Original volume = $7 \times 8 \times 5 = 280$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$;
new volume = 2240 cm³.

Exit-ticket worked solutions

1. 2560 cm³; 8 times as large

Original volume = $8 \times 8 \times 5 = 320$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 2560 cm³.

2. 2268 cm³; 27 times as large

Original volume = $2 \times 6 \times 7 = 84$. Three dimensions each scale by 3, so volume scales by $3^3 = 27$; new volume = 2268 cm³.

3. 896 cm³; 8 times as large

Original volume = $7 \times 2 \times 8 = 112$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 896 cm³.

4. 288 cm³; 8 times as large

Original volume = $3 \times 3 \times 4 = 36$. Three dimensions each scale by 2, so volume scales by $2^3 = 8$; new volume = 288 cm³.



Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Use volume formulas and cubic scale factors. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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